

False Positive Catamenial Epilepsy Classification: A Simulation Study

Research Article

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Abstract

Objective: To quantify false-positive catamenial epilepsy (CE) classification when seizure and menstrual-cycle diaries are independent.

Methods: We simulated 100,000 synthetic participants for 36 months in healthy ovulatory and heterogeneously ovulatory menstruating-age cohorts. CHOCOLATES generated seizure diaries, and HORMONE-CYCLE version 0.4.0 independently generated menstrual events and latent daily hormone envelopes; diaries were aligned by calendar day. A healthy ovulatory cohort and a heterogeneous cohort were separately evaluated. We applied standard phase definitions and hormone pattern-specific seizure-frequency ratio thresholds across calendar, complete-cycle, and full-diary windows. A negative-binomial analysis was used as a statistical calibration check.

Results: Three-month false-positive rates were 41.6% (healthy cohort) and 51.0% (heterogeneous cohort). In 36-month windows, CE was classified in 11.4% (healthy cohort) and 35.8% (heterogeneous cohort). In the heterogeneous cohort, the C1/C2 rate was 11.6%, whereas C3 occurred in 37.0% of applicable windows. C1/C2 negative-binomial rates were 4.3% and 4.3%; the exploratory C3 model was positive in 6.6% of classifiable audit participants.

Significance: False-positive CE classification is common when independent seizure and menstrual-cycle simulations are overlaid, particularly for heterogeneous cycles when C3 logic is used. Population, observation duration, phase definitions, and minimum-data rules materially affect results and should be reported explicitly.

Keywords: catamenial epilepsy; false-positive classification; menstrual cycle; seizure diary; simulation

Key Points

- Despite independence, 36-month Herzog false-positive rates were 11.4% in healthy-ovulatory and 35.8% in heterogeneous cohorts.
- The heterogeneous-cohort excess was primarily C3/inadequate-luteal-phase driven; full-diary C1/C2 rates were similar between cohorts.
- Three-month observation windows produced substantially higher false-positive classification than longer monitoring.
- Report the phase, its temporal definition, the comparison method, and prespecified minimum-data requirements.

Introduction

Catamenial epilepsy (CE) describes seizure exacerbation in relation to the menstrual cycle. Perimenstrual (C1), periovulatory (C2), and inadequate-luteal-phase (C3) patterns are the most often reported since formulated by Herzog and colleagues¹. However, reported prevalence of CE varies with phase definitions, ratio thresholds, diary duration, and eligibility rules. Herzog and colleagues reported prevalence of 39.1% using three complete cycles and a two-of-three-cycle rule, while 44.2% was reported in the NIH Progesterone Treatment Trial.^{1 2}

Seizures are sometimes temporally structured through clustering, circadian rhythms, and multidien rhythms. When these processes are independent of menstrual cycling, time windows can still align by chance. Such chance occurrences represent *false-positive classifications of CE*. Ratio definitions are especially vulnerable with sparse events: a small comparator count can yield a large or undefined ratio.³⁻⁵

Determining the rate of false-positive classification of CE under independence will contribute improved understanding of the state of the field of CE research and diagnosis. While the false positive rate will not provide an estimate of population prevalence of CE, it can contextualize the contribution of false-positive CE classification to current commonly reported CE rates. Such research is important because patients with epilepsy have treatments modified, added, or withheld based on beliefs around the presence or absence of CE.

We conducted a realistic simulation with diaries and hormone cycles generated independently to estimate false-positive CE classification across definitions and observation windows. The study evaluates specificity behavior

only. Our aim was to determine the expected “noise” produced by completely independent systems overlapping by chance.

Methods

Study design

For each synthetic participant, seizure and menstrual/hormone diaries were generated independently for 36 months and aligned directly by calendar-day index. Analyses were stratified by cohort and phase-labeling mode.

Cohorts

Two cohorts were used: a healthy cohort and a heterogeneous cohort. The healthy ovulatory cohort included 50,000 synthetic participants aged 18–45 years with regular ovulatory cycling. The heterogeneous menstruating-age cohort included 50,000 synthetic participants aged 13–55 years and allowed anovulation and irregularity. Rates of polycystic ovary syndrome, peri-menarche, perimenopause, and dysmenorrhea were sampled based on data-driven population estimates (Appendix S1).

Diary simulators

CHOCOLATES generated realistic self-reported seizure counts for all synthetic patients. The tool uses a hierarchical gamma–Poisson (equivalent to negative binomial) process with participant heterogeneity, L-relationship, clustering, multidien cycles, and maximum seizure rates.⁶ HORMONE-CYCLE version 0.4.0 is a custom simulator developed for the present project and released as an open-source community tool (see Appendix).⁷ It begins on a randomly selected day within a cycle and proceeds forward. It assigns age-specific participant variability components, samples right-skewed cycle realizations, maps every in-cycle daily source-hormone median, adds modest cycle-specific timing and relative-shape variation in a separate deterministic stream, retains the late-luteal progesterone tail, and couples fitted age-50-or-older long-cycle episodes to ovulatory status. Full hormone trajectories were retained for validation and the prespecified 1% daily audit sample. Both hormone and seizure simulators were designed in a data-driven manner to recapitulate statistical features observed within and across real patient populations, including randomized start times. The exact targets, sampling levels, realized distributions, and assumption-by-assumption rationale are reported in the Appendix.^{8–10 11}

As a separate aggregate cross-check, simulated 12-month age-stratified cycle means and participant-specific standard deviations were compared with a global period-tracking cohort.¹² The comparison was prespecified as an external summary-level check rather than individual-level validation (see Appendix S1).

Phase labeling and Herzog criteria

Strict Herzog phase labeling was the primary phase-definition scheme. The first day of a menstruation was labeled day 1. Menstrual phase comprised forward days 1–3 and backward days –3 to –1; follicular phase, forward days 4–9; ovulatory phase, forward days ≥ 10 through backward day –13; and luteal phase, backward days –12 to –4. Days outside these phases were unlabeled. A luteal-anchored sensitivity fixed the periovulatory interval at backward days –16 to –13 and assigned cycle-length variation to the follicular phase.¹⁶

Average daily seizure frequency (ADSF) was used for ratios. Windowed ratios were $C1 = \text{ADSF}(M) / \text{ADSF}(F+L)$ with threshold 1.69, $C2 = \text{ADSF}(O) / \text{ADSF}(F+L)$ with threshold 1.83, and $C3 = \text{ADSF}(O+L+M) / \text{ADSF}(F)$ with threshold 1.62 in applicable inadequate-luteal-phase cycles/windows. These exact distributional inflection values, rather than a rounded twofold rule, were used.¹⁶ A ratio was undefined when its numerator or comparator phase was absent or when both phase rates were zero; a positive numerator with zero comparator was treated as positive infinity.

C3 applicability

In the heterogeneous cohort, the software assigned a simulator-generated inadequate-luteal-phase indicator to all nonovulatory cycles and cycles without a valid ovulation day or progesterone field. For other ovulatory cycles, it used the maximum progesterone concentration on days 5–9 after ovulation, falling back to all postovulation days if that window was absent, and designated values < 5.0 ng/mL as inadequate. The C3 ratio then used only designated

cycles. The 5.0-ng/mL cutoff follows the operational ovulatory criterion in the original Herzog study (see Appendix S1).¹⁶

CE Classification definitions

The exact Herzog 2004 cycle definition required exactly three complete 23–35-day cycles and the same positive pattern in at least two cycles.¹ In contrast, our windowed Herzog cycle definition pooled the phase-specific seizure-frequency ratios within each calendar, complete-cycle, or full-diary window. The minimum-data sensitivity required at least four calendar months or six complete cycles and at least four seizure days. These pragmatic thresholds represent an investigator-selected minimum-data sensitivity intended to reduce classifications dominated by extremely sparse event counts; they were not biologically derived. Nearby event thresholds were examined without searching for an optimal cutoff (Appendix S1). Cycle reproducibility required the same positive pattern in at least two-thirds of eligible cycles and a positive pooled ratio.

Model-concordant statistical calibration checks

Recurrent heterogeneous seizure counts are overdispersed, motivating negative binomial rather than simple Poisson models.⁶ The prespecified C1/C2 check used within-participant daily-count generalized linear models with log link, menstrual and ovulatory indicators, and cycle fixed effects where estimable. One-sided Wald tests were Holm-adjusted across C1 and C2; positivity required adjusted $P < .05$ and the corresponding Herzog rate-ratio threshold. Because CHOCOLATES uses a negative-binomial-like count process, this was a model-concordant statistical calibration check.

A separately specified exploratory C3 model used the deterministic 1% daily audit sample. It restricted the heterogeneous cohort to complete cycles carrying the simulator-generated inadequate-luteal-phase indicator and contrasted combined O+L+M with F days using cycle fixed effects. Positivity required a one-sided $P < .05$ and rate ratio ≥ 1.62 . This standalone exploratory family was not pooled with C1/C2 and therefore received no cross-family multiplicity adjustment (Appendix S1).

Outcomes and statistics

The primary outcome was the participant-window false-positive classification rate among classifiable windows under independence. We also report rates among all attempted windows, explicit indeterminate reasons, and mutually exclusive pattern categories. An indeterminate window was one in which an applicable rule could not return positive or negative because required phases, cycles, events, or denominator information were insufficient; non-applicability was kept separate. Definition-by-window-by-cohort estimates with fewer than 1,000 classifiable participant-windows were flagged as unstable. Wilson intervals quantify Monte Carlo uncertainty in a simulated proportion and do not include model-form uncertainty. As a demonstration of how small studies may look, ten thousand simulated studies of 30, 50, or 100 participants sampled one random valid 3-month window per participant.

Ethics

This study used only synthetic data. Institutional review board approval, patient consent, and clinical-trial registration were not applicable.

Results

Simulation cohort

The completed run included 100,000 synthetic participants. The healthy ovulatory cohort had 100.0% ovulatory cycles by design; the heterogeneous cohort had 78.9% ovulatory cycles, greater within-participant cycle-length variability, and similar seizure burden (Table 1). Every diary began at a randomly selected menstrual-cycle phase.

Full-diary CE classification

We used strict Herzog phase labeling and the prespecified windowed Herzog pattern-specific ratio thresholds across each participant's full 36-month diary. CE was classified in 11.4% of classifiable healthy-ovulatory participant-windows and in 35.8% of classifiable heterogeneous-cohort participant-windows under independence (Table 2). In the heterogeneous cohort, the C1/C2 union was 11.6%, whereas C3 was positive in 37.0% of C3-applicable

windows. In the full 36-month diaries, 1,109 of 50,000 simulated participants in the healthy cohort (2.2%) and 1,068 of 50,000 in the heterogeneous cohort (2.1%) met both C1 and C2 criteria. In the heterogeneous cohort, the analogous combined group included 564 participants (1.1%) who did not meet C3 and 504 (1.0%) who also met C3. Thus, the between-cohort difference in the composite endpoint was primarily driven by the simulator-generated inadequate-luteal-phase designation. In the retained 1% audit sample, the heterogeneous cohort's mean inadequate-luteal-phase cycle fraction was 22.9%.

Observation windows

Three-month false-positive rates were 41.6% and 51.0% in the healthy and heterogeneous cohorts, respectively. Rates declined with longer monitoring but remained definition dependent (Figure 1). C1, C2, and C3 false-positive rates declined with longer monitoring (Figure 1). Exact Herzog applied to three complete cycles yielded 50.5%45.2% and 52.2%41.1% among classifiable windows, while many attempted windows were indeterminate. Appendix S1 reports C3 across every saved calendar and complete-cycle duration, minimum-data threshold sensitivities, and cumulative simulated C1, C2, and C3 ratio distributions.

Calibration checks

The full-diary C1/C2 negative-binomial false-positive rate was 4.3% in both healthy and heterogeneous cohorts, close to the prespecified 5% Type I error rate. In the 1% daily audit sample, 428 of 500 heterogeneous participants had a C3-applicable ratio window, but only 198 met the exploratory C3 model's four-complete-inadequate-luteal-phase-cycle and four-seizure-day requirements. 13 participants were positive (6.6%; 95% Wilson confidence interval, 3.9%–10.9%; 2.6% of all 500 attempted participants).

Study-level and feature summaries

Across 10,000 simulated 3-month studies, apparent prevalence varied widely, particularly at n=30 (Table 3 and Figure 3). Associations with seizure burden, cycle variability, ovulatory fraction, age, and configured modifiers are reported in Appendix S1 as relationships between simulator inputs or realizations and simulated classifications.

Discussion

This study quantified the false positive classification of several CE definitions using simulations that could not possibly include CE. Using pooled ratio criteria in random three-calendar-month windows, false-positive classification occurred in 41.6% of healthy-ovulatory and 51.0% of heterogeneous-cohort participant-windows. These results demonstrate the potential inaccuracy of ratio-based classification with short observation periods. They should not be compared directly with published CE prevalence estimates^{1 2} of 39.1% and 44.2%, because those estimates were obtained in clinical cohorts using different cycle-eligibility, reproducibility, and denominator rules.

The analysis most closely aligned with the original Herzog 2004 procedure applied the same-pattern-in-two-of-three rule to exactly complete three 23–35-day cycles. Under that definition, false-positive classification occurred in 50.5%45.2% and 52.2%41.1% of classifiable windows in the healthy and heterogeneous cohorts, respectively, although many attempted windows were indeterminate. Even this comparison should be interpreted cautiously because the simulation represents complete independence and does not reproduce the mixture of true hormone-associated and unrelated seizure patterns present in true clinical cohorts.

Accordingly, comparisons between simulated false-positive rates and empirical CE prevalence should use matched definitions, observation periods, eligibility requirements, reproducibility rules, and denominators. Numerical similarity alone does not establish how much of the reported clinical prevalence is attributable to false-positive classification.

The largest heterogeneous-cohort excess was C3-driven, whereas C1/C2 behavior was similar between cohorts.

Our results raise the question of whether C1/C2/C3 as reported in the literature are anything other than statistical artefact. In support of the Herzog classification and definition of CE, the simulations do not perfectly overlay onto empirical and real-world relationships between the menstrual cycle and seizures. For example, when degree of C1 exacerbation is accounted for (Supplementary Table S4) the false-positive rate underestimates prevalence. Across

cumulative C1 ratio thresholds from 1.69 to 10, the proportions reported by Herzog (2015)² exceeded the corresponding simulated heterogeneous-cohort proportions by 1.07–10.82 percentage points, equivalent to relative differences of 20%–88% (Appendix Table S4).

There is a considerable body of literature on how menstrual cycle steroidal fluctuations can physiologically exacerbate seizures². Further, there are some limited conflicting studies, some of which report that treatments that stop the fluctuations of hormones may be effective in reducing seizure burden in women reporting CE.^{16 17,18}

This simulation study suggests true rates of CE may be considerably lower than those diagnosed using the Herzog criteria^{1 2}. Furthermore, that the current commonly used definitions are very vulnerable to statistical artefact. As such, this study points to several design implications for future research into CE. First, short-window ratio classifications require caution when seizure counts are sparse. Second, C1, C2, and C3 should be reported separately. Third, classifiable and all-attempted denominators should be shown together. Fourth, it may be time to reconsider CE definitions in order to increase specificity.

Future empirical studies into CE and treatments ought to consider the risk of false-positive in considering power calculations and group assignment. Future directions for the simulation work includes comparing these simulation-based rates with empirical-null distributions derived from longitudinal paired seizure and menstrual diaries, using complementary preregistered within-person time shifts and matched cross-person pairing that preserve temporal structure, missingness, and participant-level dependence. Such work requires long-term real-world seizure and menstrual cycle diaries.

Strengths and limitations

Strict Herzog labeling was retained for historical comparability despite recognized limitations of fixed menstrual-phase assignment, particularly the expanding periovulatory interval in longer cycles. The luteal-anchored sensitivity therefore tested a fixed periovulatory definition.¹⁹

The study is limited by simulator assumptions. CHOCOLATES and HORMONE-CYCLE were calibrated to published features, but summary-level calibration and the held-out aggregate cycle cross-check do not substitute for external validation using paired real diaries. Agreement with published cycle data was strongest at ages 18–45 and remained qualified after age 50 because large app cohorts reported different variability magnitudes. HORMONE-CYCLE version 0.4.0 produces latent daily endocrine envelopes rather than literal serial serum measurements: it now uses every in-cycle daily Stricker value, includes modest timing and relative-shape dispersion, and retains the terminal progesterone tail, but remains deliberately smooth and template-dominant and omits intraday pulsatility and assay error. Only two of five published long-follicular estrogen patterns are implemented; the 25% failed-wave share is an investigator-selected sensitivity setting. Perimenopausal long intervals are now enriched for anovulation, but the simulated long-versus-ordinary association is weaker than in the source cohort. The configured modifiers are stress-test assumptions, the 5.0-ng/mL C3 rule operationalizes one historical definition, and no diary missingness was simulated. Treatment changes and long-term seizure-rate trends were outside scope and could invalidate stationarity assumptions. The study does not evaluate false negatives or adjudicated seizure types.

Retrospective self-report of CE may be less reliable compared with prospective seizure diaries²⁰. This may reflect unrecognized diary-defined patterns, incomplete reporting of seizures, perceived patterns that are in fact spurious correlation, or instability of the pattern across limited observation windows. Both methods may be valid under the right conditions.²⁰

The model-concordant negative-binomial findings address implementation-level calibration under related model assumptions. These tests don't consider clinical usefulness or superiority. C1/C2 rates near 5% are expected for tests constructed around a 5% Type I error rate; ratio rules have no analogous guarantee. The exploratory C3 estimate was imprecise and used stricter eligibility in the audit sample, therefore it should not be read as a direct replacement for the ratio-based C3 result. The purpose was to show that the C3 contrast can be calibrated.

In conclusion, CE misclassification may be common even when structured seizure and menstrual-cycle diaries are independent. Diagnostic and interventional studies should prespecify diary length, phase labeling, minimum-data thresholds, zero-denominator handling, ovulatory-status handling, and reproducibility criteria before interpreting apparent menstrual clustering as hormone-linked seizure exacerbation.

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Artificial Intelligence Use Disclosure

OpenAI Codex, a generative artificial intelligence coding and writing assistant, was used to assist with code review, software and notebook updates, consistency checking, and language editing. The authors retain responsibility for the accuracy, originality, and integrity of the work.

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Conflict of Interest

DMG has been provided speaker fees from Harvard Medical School, AAN, AES, ACNS, NNS, AI in Epilepsy and Neurology, Florida Epilepsy Alliance, and UT-Austin. He also previously has been a paid consultant for Neuro Event Labs, IDR, LivaNova, Health Advances, Duke University, Bloom Insights and Wiley. He is consulting for Optum, Xenon and Praxis. He has received grants from NIH K23NS124656, NIH R21NS142800, and the American Board of Psychiatry and Neurology. BIDMC and the Lions Club. MBW is a co-founder of, serves as a scientific advisor and consultant to, and has a personal equity interest in Beacon Biosignals. The remaining authors have no conflicts of interest to disclose.

Author Contributions

Daniel M. Goldenholz: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, writing—review and editing, funding acquisition. Wesley T. Kerr: Conceptualization, methodology, validation, writing—review and editing. Sharon Chiang: Methodology, validation, writing—review and editing. M. Brandon Westover: Methodology, software, validation, writing—review and editing. Rachael L. Sumner: Methodology, validation, writing—review and editing.

Data and Code Availability

Code, configuration, derived tabular outputs, and the run manifest are available at <https://github.com/GoldenholzLab/catamenial-epilepsy-sim>. The completed primary simulation run used master seed 20260505; its manifest records configuration values, analysis-code and configuration fingerprints, file sizes, and SHA-256 checksums. The randomized-start analysis used config_random_start_full.yaml, and the Appendix S1 tables were derived from those primary outputs.

Ethical Publication Statement

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Supporting Information

Appendix S1 contains the simulator source-to-parameter map, healthy-cycle calibration and held-out aggregate validation, hormone-waveform construction and nuanced checks, representative complete-cycle hormone envelopes with explicit bleeding days, assumption review, C3 algorithm, cumulative simulated Herzog-ratio panels, window and minimum-data sensitivities, exploratory C3 negative-binomial calibration check, pattern decomposition, feature associations, Appendix Tables A1–A8, Appendix Figures A1–A5, Tables S1–S8, and Figures S1–S5.

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Tables

Table 1. Simulated cohort characteristics.

Cohort	N	Age, y Mean	Age, y SD	Cycle length, d Mean	Cycle length, d SD	Ovulatory cycles	Seizure days/month	Seizures/month
Healthy ovulatory	50,000	31.5	7.8	28.75	3.93	100.0%	2.46	6.83
Heterogeneous menstruating- age	50,000	34.0	12.1	30.09	5.61	78.9%	2.45	6.80

Note. Values are cohort-level means from the completed run. Cycle-length SD is the mean within-participant cycle-length SD. N denotes simulated participants; y, years; d, days.

355 **Table 2. Participant-window false-positive results.**

Cohort	Analysis window	Definition	Classifiable / attempted	Classifiable, % (95% Wilson CI)	All attempted, %	Indeterminate, %
Healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.4 (11.1–11.6)	11.3	0.8
		Windowed Herzog C1/C2 union	49,605 / 50,000	11.4 (11.1–11.6)	11.3	0.8
		Windowed Herzog C1	49,455 / 50,000	7.9 (7.7–8.1)	7.8	1.1
		Windowed Herzog C2	49,513 / 50,000	5.7 (5.5–5.9)	5.7	1.0
		Herzog C1/C2 + ≥ 4 seizure-day minimum	48,359 / 50,000	10.3 (10.1–10.6)	10.0	3.3
		Negative-binomial C1/C2 calibration	48,359 / 50,000	4.3 (4.1–4.5)	4.2	3.3
	3-month calendar	Windowed Herzog thresholds	45,321 / 50,000	41.6 (41.2–42.1)	37.7	9.4
	3 complete cycles	Exact Herzog 2004, any pattern	22,113 / 50,000 18,803 / 50,000	50.5 (49.9–51.2) 45.2 (44.5–45.9)	22.3 17.0	55.8 62.4
Heterogeneous menstruating-age	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	35.8 (35.4–36.3)	35.5	0.8
		Windowed Herzog C1/C2 union	49,587 / 50,000	11.6 (11.3–11.9)	11.5	0.8
		Windowed Herzog C1	49,400 / 50,000	8.1 (7.9–8.4)	8.1	1.2
		Windowed Herzog C2	49,482 / 50,000	5.6 (5.4–5.8)	5.6	1.0
		Windowed Herzog C3 only	38,583 / 50,000	37.0 (36.6–37.5)	28.6	22.8
		Herzog C1/C2 + ≥ 4 seizure-day minimum	48,333 / 50,000	10.5 (10.2–10.8)	10.1	3.3
		Negative-binomial C1/C2 calibration	48,333 / 50,000	4.3 (4.1–4.5)	4.1	3.3
	3-month calendar	Windowed Herzog thresholds	45,194 / 50,000	51.0 (50.5–51.4)	46.1	9.6
	3 complete cycles	Exact Herzog 2004, any pattern	18,153 / 50,000 15,291 / 50,000	52.2 (51.5–53.0) 41.1 (40.3–41.9)	19.0 12.6	63.7 69.4

Note. All rows use strict Herzog phase labeling. C3 is not applicable in the healthy ovulatory cohort, so the prior healthy C3 row is omitted. Indeterminate means that an applicable rule could not return positive or negative because required phases, cycles, events, or denominator information were insufficient; nonapplicability is distinct. Definition-by-window-by-cohort estimates based on fewer than 1,000 classifiable participant-windows were flagged as unstable. [C1, C2, and C3 rows evaluate each criterion separately and include participants who met another criterion. Classifiable denominators may differ between criteria.](#)

363 **Table 3. Study-level Monte Carlo summaries for 3-month windows.**

Cohort	N/study	Definition	Mean apparent prevalence	2.5th–97.5th percentiles	Pr(apparent prevalence $\geq 39.1\%$)	Pr(apparent prevalence $\geq 44.2\%$)
Healthy ovulatory	30	Windowed Herzog thresholds	37.7%	20.0%–56.7%	46.9%	20.3%
	30	Windowed Herzog C1/C2 union	37.7%	20.0%–56.7%	46.9%	20.3%
	50	Windowed Herzog thresholds	37.8%	24.0%–52.0%	42.3%	15.3%
	50	Windowed Herzog C1/C2 union	37.8%	24.0%–52.0%	42.3%	15.3%
	100	Windowed Herzog thresholds	37.5%	28.0%–47.0%	34.0%	7.8%
	100	Windowed Herzog C1/C2 union	37.5%	28.0%–47.0%	34.0%	7.8%
Heterogeneous menstruating-age	30	Windowed Herzog thresholds	46.0%	30.0%–63.3%	80.1%	53.8%
	30	Windowed Herzog C1/C2 union	38.0%	20.0%–56.7%	47.5%	21.5%
	50	Windowed Herzog thresholds	46.0%	32.0%–60.0%	84.3%	56.0%
	50	Windowed Herzog C1/C2 union	38.0%	24.0%–52.0%	43.4%	15.5%
	100	Windowed Herzog thresholds	46.1%	36.0%–56.0%	90.6%	62.1%
	100	Windowed Herzog C1/C2 union	38.1%	29.0%–48.0%	38.6%	9.3%

364 **Note.** Each row summarizes 10,000 simulated studies using one random valid 3-month window per selected participant. Ranges are empirical
365 percentiles across simulated studies, not confidence intervals. The 39.1% benchmark is aligned with the Herzog 2004 three-cycle/two-of-three rule;
366 44.2% is the reported NIH Progesterone Trial proportion.
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Figure Legends and Figures

Figure 1. False-positive rates by length of observation window. Bars show the percentage of classifiable calendar-window participant windows classified by strict Herzog phase labeling under independence. The dashed 5% line is a visual reference to the prespecified regression Type I error rate; it is not an expected target for the Herzog ratio rules. **Figure 1. False-positive C1, C2, and C3 classification by observation duration.** Grouped bars show each criterion separately in healthy ovulatory participants, panel A, and the heterogeneous cohort, panel B, under independence and strict Herzog phase labeling. Each duration includes 50,000 attempted calendar-window participant-windows per cohort. Rates use each criterion's classifiable denominator, and error bars show 95% Wilson confidence intervals. Participants may meet more than one criterion, so the bars are not mutually exclusive. C3 is not applicable in the healthy ovulatory cohort.

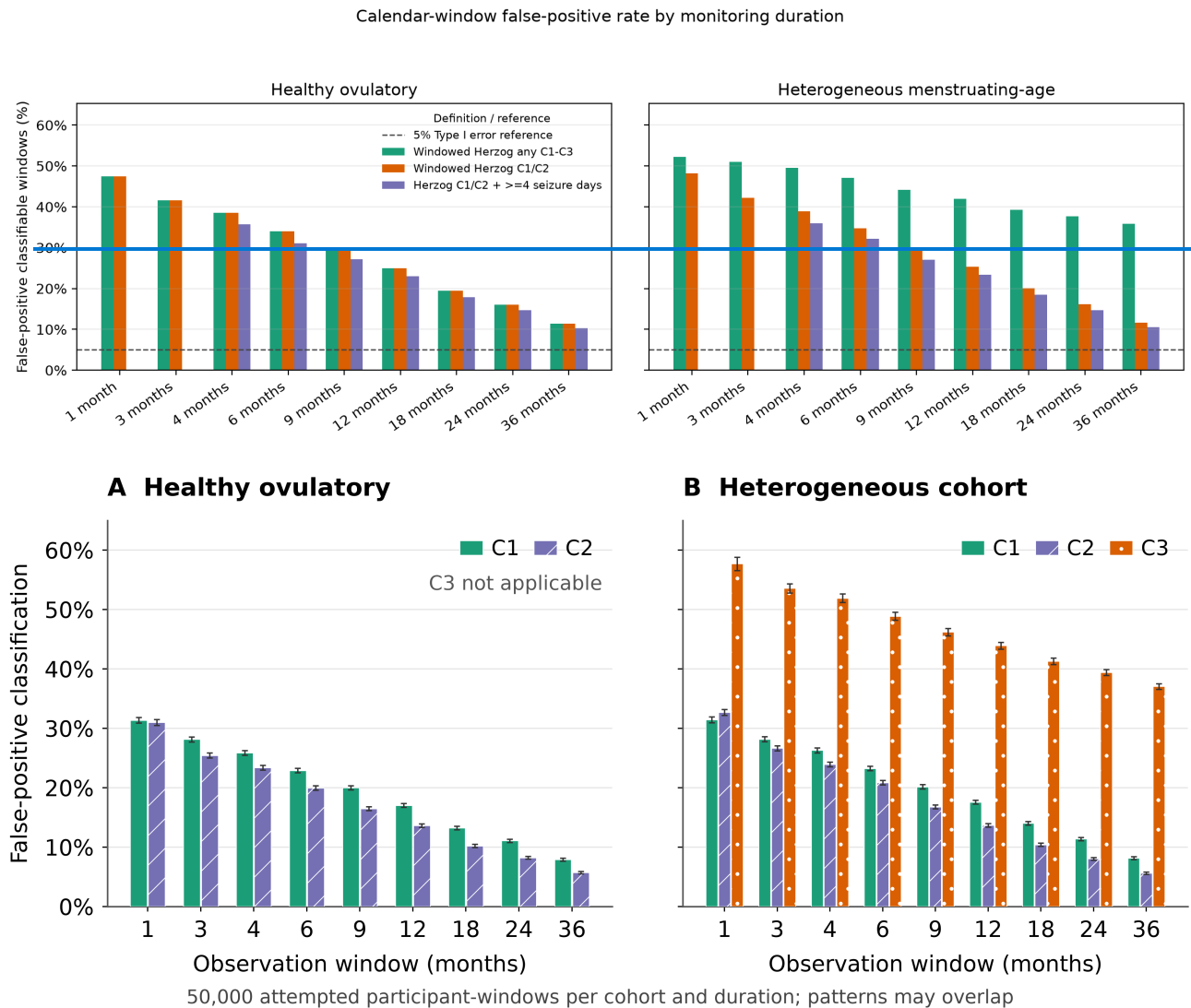


Figure 2. Pattern decomposition of false-positive catamenial classifications. Stacked bars show mutually exclusive participant-window categories in 36-month strict-Herzog windows. Percentages use all attempted windows. C3 is not applicable in the

healthy ovulatory cohort; zero-height C3 segments there are not measured zero rates. Note the C1+C2 category excludes C3, so its 1.1% heterogeneous-cohort segment is not the total C1+C2 overlap.

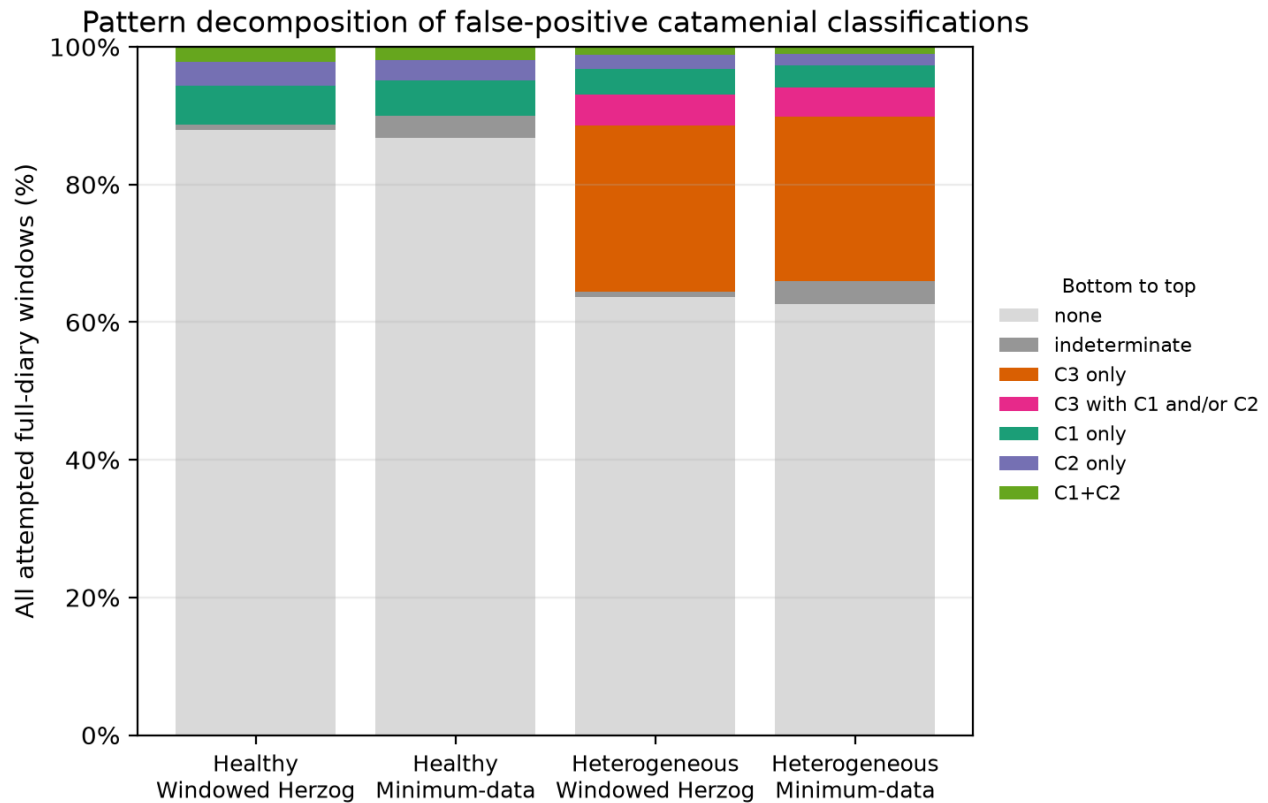


Figure 3. False-positive prevalence in simulated 3-month studies. Distributions show apparent prevalence in 10,000 studies using strict-Herzog windowed thresholds for $n=30$, 50, and 100. Vertical lines mark the historical 39.1% and 44.2% benchmarks.

